Deep Learning-Based Super-Resolution of WRF 2 m Temperature Fields

Jakub Siedlecki

AGH University of Krakow

Applied Machine Learning

Objective

Task: controlled super-resolution of WRF 2-metre temperature fields.

The model receives a synthetically degraded and interpolated field. The target is the original high-resolution WRF field:

$$X_{\text{raw}} \to Y_{\text{raw}}, \qquad \widehat{Y} \approx Y_{\text{raw}}$$

Research question

Can a U-Net recover high-resolution T2 structure better than the interpolation baseline?

Key assumption: the baseline already preserves most large-scale temperature patterns. The network should learn the missing local correction.

Experiment

Field	Value
Variable	WRF `T2`
Period	January 2026
Leads	0–23 h
Domain	319×319
Degradation	$\times 4$ averaging
Patch size	64×64
Stride	32
Test patches	11,664

Data pipeline

Controlled degradation

Original high-resolution WRF T2 is degraded and then upsampled back to the WRF grid.

$$Y_{\text{raw}} \xrightarrow{\times 4 \text{ averaging}} Y_{\text{LR}} \xrightarrow{\text{upsampling}} X_{\text{raw}}$$

The model learns the inverse mapping:

$$X_{\text{raw}} \xrightarrow{\text{U-Net}} \widehat{Y} \approx Y_{\text{raw}}$$

Patch dataset

Patch size	64×64
Stride	32 grid cells
Split	by initialization
Test set	11,664 patches

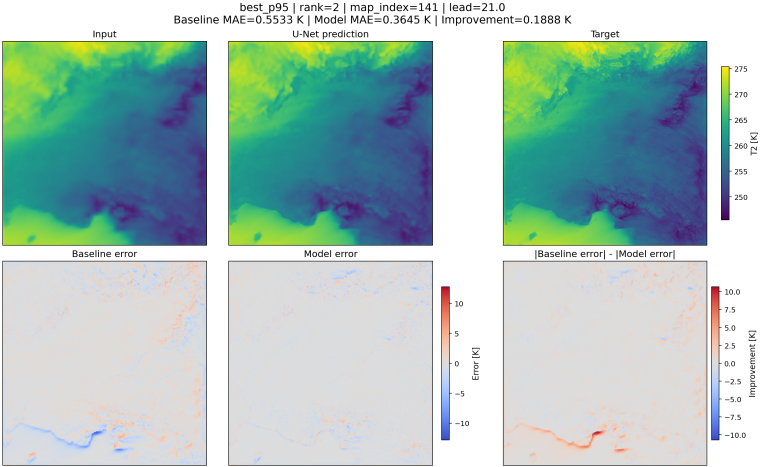
Why split by initialization?

It avoids placing highly related patches from the same forecast case in both train and test sets.

Important detail

The baseline is the upsampled degraded field. It already contains most of the large-scale temperature structure.

Full-domain example



Selected full-domain case. Top row: input, model prediction, target. Bottom row: baseline error, model error, and error difference.

Why U-Net?

Image-to-image reconstruction

U-Net is designed for tasks where input and output share the same spatial grid.

Here:

$$64 \times 64 \text{ T2} \longrightarrow 64 \times 64 \text{ T2}$$

The task is therefore **spatial reconstruction**, not classification.

Architectural intuition

- encoder: broader spatial context,
- decoder: high-resolution output,
- skip connections: local detail.

Meteorological motivation

T2 fields combine two types of structure:

Large scale smooth temperature patterns

Local scale gradients modified by terrain and surface effects

U-Net is useful because it can combine both in one encoder–decoder model.

Main intuition

The model should preserve spatial coherence while recovering local residual structure.

Model architecture

Shape flow

Stage	Shape
Input	$1 \times 64 \times 64$
Encoder 1	$32 \times 64 \times 64$
Downsample	$32 \times 32 \times 32$
Encoder 2	$64 \times 32 \times 32$
Downsample	$64 \times 16 \times 16$
Bottleneck	$128 \times 16 \times 16$
Decoder 1	$64 \times 32 \times 32$
Decoder 2	$32 \times 64 \times 64$
Output	$1 \times 64 \times 64$

Final configuration

Backbone	U-Net
Levels	2 downsampling stages
Base filters	32
Activation	SiLU
Output	one T2 channel
Parameters	470,977

Residual output

$$\widehat{Y} = X + f_\theta(X)$$

The network predicts the correction to the interpolated baseline.

Practical meaning

Since the interpolated input already contains the large-scale field, the model can focus on small-scale reconstruction errors.

Residual learning and loss

Residual formulation

The model predicts a correction to the interpolated baseline:

$$\widehat{Y} = X + f_\theta(X)$$

This is suitable because X is already a strong approximation of Y.

Evaluation metrics

- **MAE**: mean absolute error.
- **RMSE**: large-error-sensitive metric.
- **Bias**: signed mean error.
- **Gradient MAE**: spatial-detail metric.

Training loss

$$\mathcal{L} = \mathrm{MAE}(Y, \widehat{Y}) + \lambda\,\mathrm{MAE}(\nabla Y, \nabla\widehat{Y})$$

$$\lambda = 0.1$$

$$\mathrm{MAE} = \frac{1}{HW}\sum_{i,j} |Y_{ij} - \widehat{Y}_{ij}|$$

$$\mathrm{GMAE} = \mathrm{MAE}(\nabla Y, \nabla\widehat{Y})$$

Global test-set results

Metrics

Metric	Baseline	U-Net
MAE [K]	0.3261	**0.2122**
RMSE [K]	0.5386	**0.3610**
Bias [K]	0.0002	-0.0150
Grad. MAE	0.1794	**0.1659**

Improvement

- MAE: **34.9%** lower.
- RMSE: **33.0%** lower.
- Gradient MAE: **7.5%** lower.

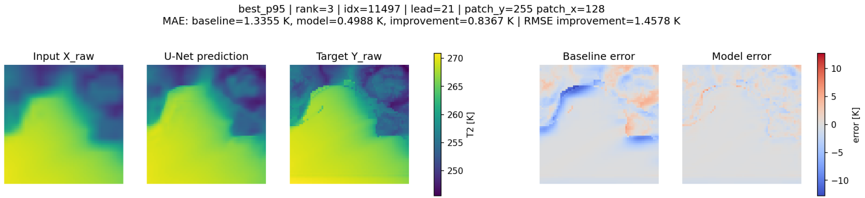
Patch-level robustness

Metric	Median gain	Improved
MAE	0.0823 K	95.3%
RMSE	0.1037 K	93.8%
Grad. MAE	0.0097	91.9%

Interpretation

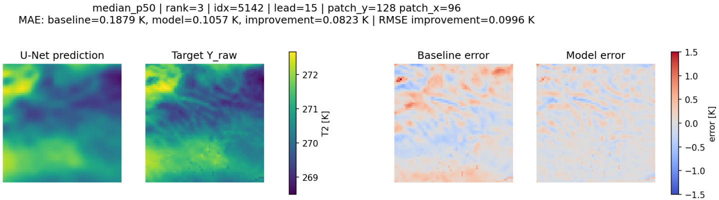
The model improves pointwise reconstruction strongly. Fine-scale gradients improve less, so local structure remains the harder part of the task.

Best-case patch



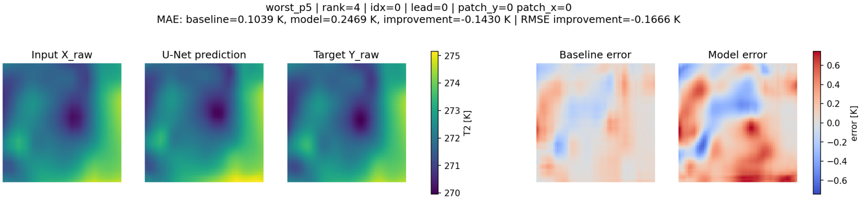
High-improvement patch. The model substantially reduces the local reconstruction error relative to the interpolation baseline.

Median-case patch



Representative patch close to the median improvement level. This is useful for judging typical model behaviour.

Worst-case patch



Low-improvement or failure-case patch. This highlights where the model may still smooth or misplace local spatial structure.

Conclusion

Main conclusion

A residual U-Net improves the interpolation baseline for controlled WRF T2 super-resolution.

What worked

- residual learning,
- SiLU activation,
- gradient-consistency term,
- initialization-based split.

Open issue

Pointwise errors improve strongly, but physically realistic fine-scale gradients remain challenging.

Limitations

- January 2026 only,
- T2 only,
- synthetic degradation,
- overlapping patches.

Next steps

- add orography / HGT,
- include more seasons,
- test additional predictors,
- explore probabilistic losses.